

Notes: Fisman & Miguel (JPE, 2007)
**Corruption, Norms, and Legal Enforcement: Evidence from Diplomatic
Parking Tickets**

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1 Big picture

- **Question.** Do corruption norms persist when officials operate outside their home institutions, and how much does formal legal enforcement matter relative to norms?
- **Setting.** United Nations diplomats in New York City. Before November 2002, diplomatic immunity made unpaid parking violations essentially legally unenforced; after the policy change, repeat violators could lose diplomatic license plates.
- **Outcome.** Unpaid New York City parking violations by country and by individual diplomat-month.
- **Main norms result.** Diplomats from countries with higher measured corruption accumulated substantially more unpaid parking violations when enforcement was absent.
- **Main enforcement result.** After New York City introduced credible enforcement, violations fell by more than 98 percent on average.
- **Core takeaway.** Both norms and enforcement matter. Culture predicts behavior even far from home, but credible punishment can dramatically change behavior.

2 Incremental contribution of this paper

- **Revealed-preference corruption measure.** The paper moves beyond perception surveys by building a behavioral measure of corruption: actual rule breaking by public officials when punishment is nearly absent.
- **Clean separation of norms from enforcement.** Diplomats from many countries face the same city, the same parking rules, and the same near-zero legal punishment before 2002.
- **Direct enforcement shock.** The 2002 license-plate confiscation policy creates a sharp shift from weak to credible enforcement in the same population.
- **Comparable cross-country measure.** The paper creates a country-level corruption measure for 149 countries based on identical behavior in the same host environment.
- **Individual-level dynamics.** Matching Blue Book personnel records to violations lets the authors examine how behavior evolves with diplomats' time in New York.

3 Data and measurement

3.1 Parking violations data

Data come from the New York City Department of Finance and cover unpaid diplomatic parking violations from November 24, 1997 to November 21, 2005. The pre-enforcement period runs from November 1997 through October 2002; the post-enforcement period begins in November 2002.

Interpretation. The outcome is a real behavior that resembles corruption: abusing an entrusted legal privilege for private benefit.

3.2 Diplomat, mission, and country data

The authors use UN Blue Books from 1997 to 2005 to count mission staff and track diplomat tenure. The main corruption measure is a 1998 country corruption index from Kaufmann, Kraay, and Mastruzzi, rescaled so higher values denote more corruption.

Interpretation. Mission size, vehicle counts, country income, and region fixed effects help distinguish corruption norms from opportunity and resources.

4 Models and empirical specifications

4.1 Country-level negative binomial model

$$E[Violations_{it} | X] = \exp(\alpha + \beta_1 Corruption_i + \beta_2 Enforcement_t + \beta_3 Diplomats_i + X_i' \gamma).$$

There are two observations per country: pre-enforcement and post-enforcement. A negative binomial model is used because the count outcome is overdispersed.

Interpretation. β_1 measures the culture/norm component under a common host-city environment; β_2 measures the effect of the enforcement regime.

4.2 Diplomat-level tenure model

$$E[Violations_{imt} | X] = \exp(\alpha + \beta_1 Corruption_i + \beta_2 \log Tenure_{mt} + \beta_3 Corruption_i \times \log Tenure_{mt} + Month_t + controls).$$

Interpretation. Positive tenure effects imply diplomats learn that violations are not punished; the interaction shows whether this learning differs by home-country corruption.

5 Identification and empirical design

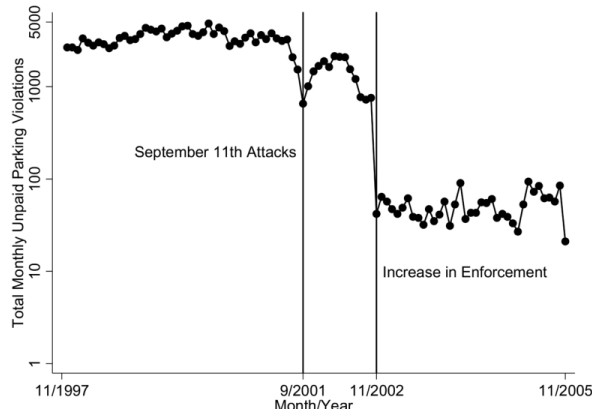
The design uses a common legal environment, near-zero enforcement before 2002, a credible enforcement shock after 2002, a relatively homogeneous occupational group, and robustness checks using income, regions, mission size, vehicles, distance, trade, aid, and alternative outcomes. The setting is unusual, so the results are best read as evidence on the interaction between norms and enforcement rather than a complete model of all corruption.

6 Tables and figures

6.1 Figure 1: Total monthly New York City parking violations by diplomats, 1997–2005 and Table 2: Descriptive statistics

Read: Figure 1 shows thousands of monthly violations before enforcement and a dramatic collapse after the 2002 enforcement increase. Table 2 shows mean country-level unpaid violations falling from 977.9 pre-enforcement to 11.3 post-enforcement.

Economic message. The figure and summary statistics establish the key contrast: norms drive cross-country variation in a low-enforcement regime, while credible enforcement nearly eliminates the behavior.



(a) Monthly violations and enforcement shift

TABLE 2
DESCRIPTIVE STATISTICS

Variable	Mean	Standard Deviation	Observations
A. Country-Level Data			
Unpaid New York City parking violations: ^a			
11/1997–11/2002	977.9	2,000.9	149
11/2002–11/2005	11.3	18.4	149
Unpaid and paid New York City parking violations, 11/1997–11/2002 ^a	1,066.2	2,021.4	149
After-hours New York city parking violations, 11/1997–11/2002 ^a	40.6	72.7	149
Diplomats in the country UN mission, 1998 ^b	11.8	11.1	149
Number of license plates registered to the country's UN mission, 2006 ^c	10.5	14.0	139
Country corruption index, 1998 ^d	.01	1.01	149
Log per capita income (1998 US\$) ^e	7.35	1.59	149
Average government wage/country per capita income, early 1990s ^f	2.83	2.38	92
Log weighted distance between populations ^g	9.12	.41	149
Log total trade with the United States (1998 US\$) ^h	20.3	2.7	146
Received U.S. economic aid (indicator), 1998 ⁱ	.69	.46	147
Received U.S. military aid (indicator), 1998 ⁱ	.63	.49	147
B. Diplomat-Level Data			
Monthly unpaid New York City parking violations: ^a			
11/1997–11/2002	.90	3.30	25,123
11/2002–11/2005	.02	.19	15,806
Length of time at the UN mission in New York City (in months) ^h	7.7	12.4	40,929

^a Source: New York City, Parking Violations Database (provided to the authors by the New York City Department of Finance).
^b Source: United Nations Blue Books, 1997–2005.
^c Source: U.S. Department of State Office of Foreign Missions (provided to the authors by Deputy Director Murray Smith).
^d Composite index from Kaufmann et al. (2005), but here higher values indicate more corruption.
^e Source: World Development Indicators (2005).
^f Source: Schiavo-Campo et al. (1999); exact year differs by country.
^g Source: Mayer and Ziguaga (2005).

(b) Descriptive statistics

6.2 Table 1: Average unpaid annual New York City parking violations per diplomat

Read: Table 1 ranks countries by pre-enforcement violations per diplomat. The highest-violation countries include Kuwait, Egypt, Chad, Sudan, Bulgaria, Mozambique, Albania, Angola, Senegal, and Pakistan. The bottom of the ranking contains countries with zero or near-zero unpaid violations.

Economic message. The country ranking makes the revealed-preference measure concrete: some missions generated hundreds of violations per diplomat while others generated almost none under the same institutional environment.

TABLE 1
AVERAGE UNPAID ANNUAL NEW YORK CITY PARKING VIOLATIONS PER DIPLOMAT, NOVEMBER 1997 TO NOVEMBER 2005

Parking Violations Rank	Country Name	Violations per Diplomat, Pre-enforcement (11/1997–11/2002)	Violations per Diplomat, Post-enforcement (11/2002–11/2005)	UN Mission Diplomats in 1998	Corruption Index, 1998	Country Code
1	Kuwait	249.4	.15	9	–1.07	KWT
2	Egypt	141.4	.33	24	.25	EGY
3	Chad	125.9	.00	2	.84	TCD
4	Sudan	120.6	.37	7	.75	SDN
5	Bulgaria	119.0	1.64	6	.50	BGR
6	Mozambique	112.1	.07	5	.77	MOZ
7	Albania	85.5	1.85	3	.92	ALB
8	Angola	82.7	1.71	9	1.05	AGO
9	Senegal	80.2	.21	11	.45	SEN
10	Pakistan	70.3	1.21	13	.76	PAK
11	Ivory Coast	68.0	.46	10	.35	CIV
12	Zambia	61.2	.15	9	.56	ZMB
13	Morocco	60.8	.40	17	.10	MAR
14	Ethiopia	60.4	.62	10	.25	ETH
15	Nigeria	59.4	.44	25	1.01	NGA
16	Syria	53.3	1.36	12	.58	SYR
17	Benin	50.4	6.50	8	.76	BEN
18	Zimbabwe	46.2	.86	14	.13	ZWE
19	Cameroon	44.1	2.86	8	1.11	CMR
20	Montenegro and Serbia	38.5	.05	6	.97	YUG
21	Bahrain	38.2	.65	7	–.41	BHR
22	Burundi	38.2	.11	3	.80	BDI

(a) Highest pre-enforcement violators

TABLE 1
(Continued)

Parking Violations Rank	Country Name	Violations per Diplomat, Pre-enforcement (11/1997–11/2002)	Violations per Diplomat, Post-enforcement (11/2002–11/2005)	UN Mission Diplomats in 1998	Corruption Index, 1998	Country Code
143	Japan	.0	.01	47	–1.16	JPN
144	Latvia	.0	.00	5	.10	LVA
145	Norway	.0	.00	12	–2.35	NOR
146	Oman	.0	.20	5	–.89	OMN
147	Panama	.0	.00	8	.28	PAN
148	Sweden	.0	.00	19	–2.53	SWE
149	Turkey	.0	.00	25	.01	TUR

Note.—The corruption index is from Kaufmann et al. (2005). A higher score in the corruption index denotes more corruption.

(b) Lowest pre-enforcement violators

6.3 Table 3: Country characteristics and unpaid New York City parking violations

Read: The corruption coefficient is positive and significant; the post-enforcement coefficient is about -4.4 , implying an approximate 98 percent fall in expected violations.

Economic message. Table 3 is the central regression evidence: home-country corruption predicts rule breaking when enforcement is weak, and enforcement dominates once credible sanctions arrive.

	DEPENDENT VARIABLE: UNPAID PARKING VIOLATIONS				
	(1)	(2)	(3)	(4)	(5)
Country corruption index, 1998	.48*** (.18)	.57*** (.22)	.57*** (.21)	.56** (.28)	.57* (.30)
Postenforcement period indicator (post-11/2002)	-4.41*** (.21)	-4.41*** (.21)	-4.21*** (.13)	-4.43*** (.20)	-4.41*** (.21)
Country corruption index $\times$ postenforcement period					-.01 (.28)
Diplomats	.05** (.02)	.04** (.02)	.05*** (.02)	.05** (.02)	.04** (.02)
Log per capita income (1998 US\$)		.06 (.14)	.09 (.14)	64.2* (36.9)	.06 (.14)
Africa region indicator variable			2.86*** (.48)		
Asia region indicator variable			1.99*** (.50)		
Europe region indicator variable			2.24*** (.55)		
Latin America region indicator variable			1.67*** (.56)		
Middle East region indicator variable			3.23*** (.60)		
Oceania region indicator variable			1.51** (.64)		
Log per capita income (1998 US\$) polynomials (quadratic, cubic, quartic)	No	No	No	Yes	No
Observations	298	298	298	298	298
Log pseudolikelihood	-1,570.21	-1,570.07	-1,547.69	-1,567.56	-1,570.07

NOTE.—Negative binomial regressions. White robust standard errors are in parentheses. Disturbance terms are clustered by country (there are two observations per country: pre-enforcement and postenforcement). The omitted region category is North America/Caribbean.

* Statistically significantly different from zero at 90 percent confidence.

** Statistically significantly different from zero at 95 percent confidence.

*** Statistically significantly different from zero at 99 percent confidence.

Main country-level negative binomial regressions

6.4 Figures 2 and 3: Country corruption and unpaid violations before and after enforcement

Read: Figure 2 shows a clear positive pre-enforcement relationship between corruption and log violations per diplomat. Figure 3 shows the same relationship after enforcement, but at much lower violation levels.

Economic message. The scatterplots show both forces at once: culture produces cross-country differences, while enforcement shifts the whole distribution downward.

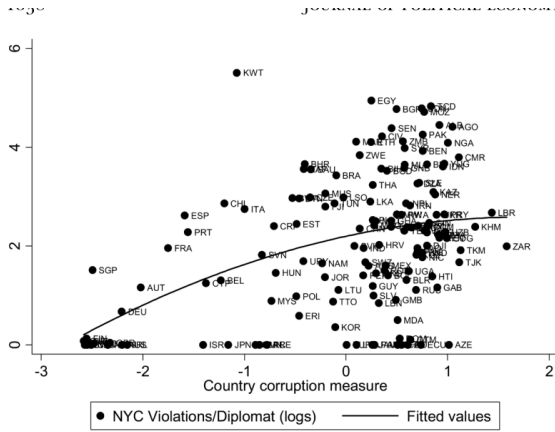


FIG. 2.—Country corruption and unpaid New York City parking violations per diplomat (in logs), pre-enforcement (November 1997 to November 2002). Country abbreviations are presented in table 1. The line is the quadratic regression fit. The x-axis is $\log(1 +$

(a) Pre-enforcement relationship

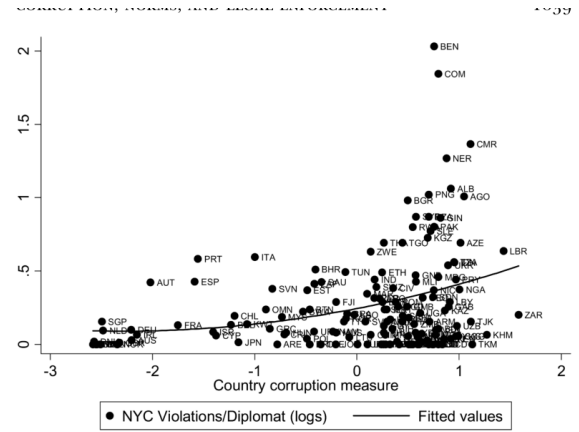


FIG. 3.—Country corruption and unpaid New York City parking violations per diplomat (in logs), postenforcement (November 2002 to November 2005). Country abbreviations are presented in table 1. The line is the quadratic regression fit. The x-axis is $\log(1 +$

(b) Post-enforcement relationship

6.5 Table 4: Sensitivity tests

Read: Table 4 tests alternative outcomes and controls, including diplomatic vehicles, paid plus unpaid violations, after-hours violations, log outcomes, OLS specifications, distance, trade, and aid.

Economic message. The robustness table narrows alternative explanations: the main result is not simply income, mission size, vehicle availability, distance, trade, or a particular count specification.

TABLE 4
COUNTRY CHARACTERISTICS AND UNPAID NEW YORK CITY PARKING VIOLATIONS, NOVEMBER 1997 TO NOVEMBER 2005:
SENSITIVITY TESTS

	DEPENDENT VARIABLE						
	Unpaid Parking Violations Negative Binomial (1)	Unpaid Parking Violations Negative Binomial (2)	Paid and Unpaid After-Hours Parking Violations Negative Binomial (3)	Log(1 + Unpaid Parking Violations) Negative Binomial (4)	Unpaid Parking Violations Negative Binomial (5)	Unpaid Parking Violations Negative Binomial (6)	Unpaid Parking Violations Negative Binomial (7)
Country corruption index, 1998	1.01*** (.32)	.48** (.24)	.47** (.18)	.56* (.30)	.37** (.16)	123.9* (72.5)	74*** (25)
Postenforcement period indicator (post-11/2002)	-4.06*** (.15)	-4.31*** (.19)	-3.36*** (.16)	-3.52*** (.21)	-2.69*** (.14)	-966.6*** (164.6)	-4.34*** (.19)
Diplomats	.06** (.03)	.01 (.02)	.05*** (.02)	.04* (.02)	.21.1* (11.2)	.05 (.03)	.05 (.03)
Log per capita income (1998 US\$)	.32 (.20)	.01 (.14)	.06 (.13)	-.02 (.17)	-.24** (.10)	-40.0 (68.4)	.02 (.17)
Average government wage/country per capita income	.15** (.06)						

(a) Robustness, part 1

UNPAID PARKING VIOLATIONS

	184	278	298	298	298	298
Log diplomats					.72*** (.15)	1.23*** (.30)
Log weighted distance of population from United States						.04 (.07)
Log total trade with the United States						-.60** (.30)
Received U.S. economic aid						.10 (.23)
Received U.S. military aid						.13 (.15)
Observations	184	278	298	298	298	298
Log pseudolikelihood	-967.23	-1,463.60	-1,816.45	-831.14	...	-1,510.79
R ²	...	...	...	...	.52	.13

NOTE.—White robust standard errors are in parentheses. Disturbances are clustered by country.
* Statistically significantly different from zero at 90 percent confidence.
** Statistically significantly different from zero at 95 percent confidence.
*** Statistically significantly different from zero at 99 percent confidence.

(b) Robustness, part 2

6.6 Table 5: Diplomat-level violations

Read: Log tenure in New York is positive and significant, and country corruption remains important. The tenure interaction implies proportional increases are strongest from a low base among diplomats from low-corruption countries.

Economic message. Norms are persistent but not immutable: exposure to a low-enforcement environment can erode rule-following behavior.

TABLE 5
UNPAID PARKING VIOLATIONS AT THE DIPLOMAT LEVEL, NOVEMBER 1997 TO
NOVEMBER 2005

	DEPENDENT VARIABLE: UNPAID PARKING VIOLA- TIONS (Monthly)	
	Negative Binomial (1)	Negative Binomial (2)
Country corruption index, 1998	.150 (.120)	.390*** (.117)
Log length of time in New York City (in months)	.084*** (.005)	.090*** (.006)
Log length of time in New York City $\times$ coun- try corruption index		-.027*** (.006)
Month fixed effects	Yes	Yes
Observations (diplomats)	40,929 (5,338)	40,929 (5,338)
Log pseudolikelihood	-23,733	-23,621

NOTE.—White robust standard errors are in parentheses. Disturbance terms are clustered by country. Observations are at the diplomat-month level. Month fixed effects are included in all regressions (thus the postenforcement indicator is not included). The log per capita income (1998 US\$) term is included as a control in cols. 1–2 (results not shown).

* Statistically significantly different from zero at 90 percent confidence.

** Statistically significantly different from zero at 95 percent confidence.

Diplomat-level negative binomial regressions

7 Short summary

- The paper studies corruption norms and enforcement using diplomatic parking violations in New York City.
- Countries with higher corruption scores generated more unpaid violations per diplomat.
- The 2002 enforcement change reduced unpaid violations by more than 98 percent.
- The result is robust to income, region, mission size, vehicles, paid violations, after-hours violations, distance, trade, and aid controls.
- Individual evidence shows violations rise with time in New York, suggesting adaptation to low enforcement.

8 Conclusion

8.1 Core conclusion

Fisman and Miguel show that corruption norms travel with officials. When diplomats from many countries operate in the same city under near-zero legal enforcement, their unpaid parking violations are strongly related to corruption in their home countries. Credible legal enforcement then produces an immediate and dramatic reduction in violations.

8.2 Incremental contribution

The paper adds a revealed-preference, field-based measure of corruption to a literature dominated by perception indices and cross-country institutional comparisons. It also provides a rare setting where cultural norms and formal enforcement can be separately observed.

8.3 Mechanism and policy implication

The mechanism is a combination of internalized norms and external incentives. In the absence of punishment, diplomats differ in willingness to exploit diplomatic immunity. When punishment becomes credible, even

diplomats from high-corruption countries largely stop violating. Anticorruption policy should therefore treat norms and enforcement as complements.

8.4 Limitation and takeaway

The setting is unusual and the behavior is parking violations rather than bribery or procurement theft. Still, the paper's clean comparison of similar officials in a common legal environment makes the evidence unusually sharp. The enduring lesson is that officials bring norms with them, but sanctions matter.